Nucleic Acids
Lecturer 2

Presented By

Dr. Ahmed Talaat Mora

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Chapter II
Nucleotides

Chapter Two: Nucleotides

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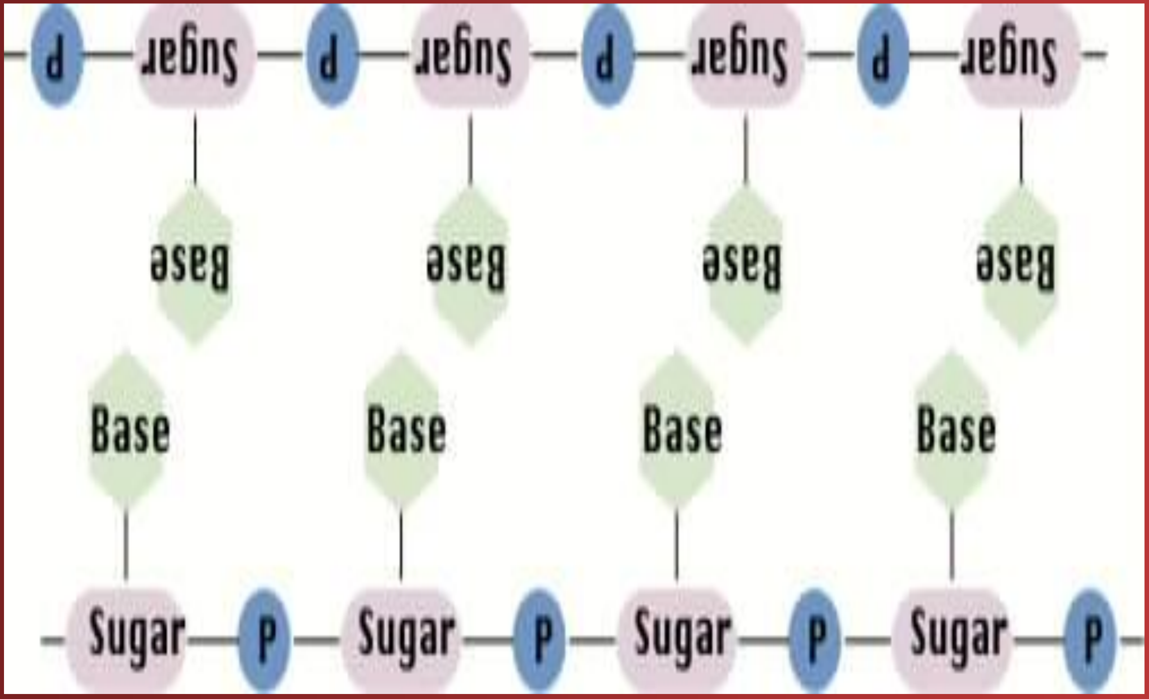
1- Nucleotides definition

- Both DNA and RNA are high-molecular-weight polymeric compounds.
- The chain-like macromolecule is made up of strings of monomeric units called nucleotides.



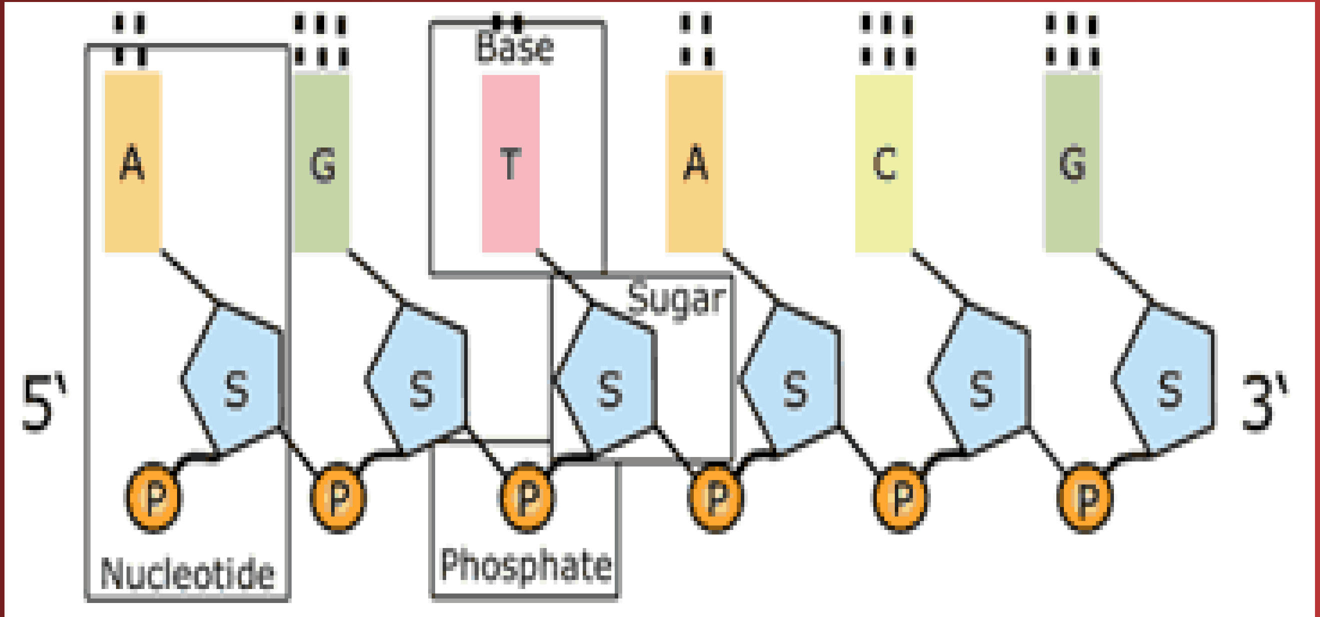
2-Nucleotide Components

- Each nucleotide is composed of three monomeric components:

a- Pentose and Deoxypentose sugar

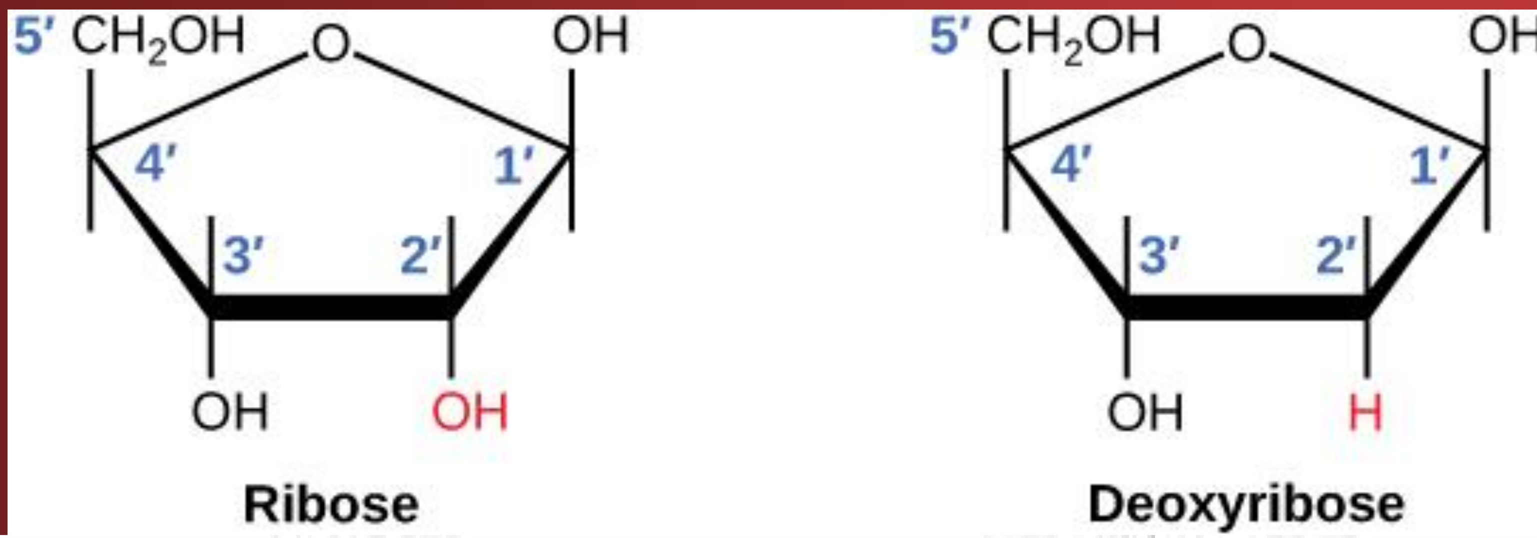
b- Nitrogenous bases

c- Phosphate group



a- Pentose and Deoxypentose sugar

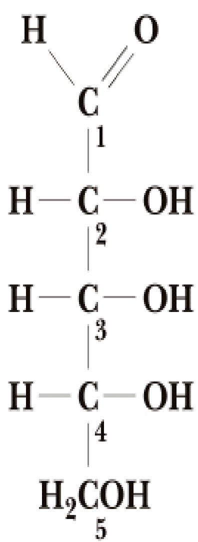
- A cyclic 5 carbon sugar, these sugars in polynucleotides occur in the D-furanose form either D-ribose in RNA or 2'-deoxyribose in DNA.



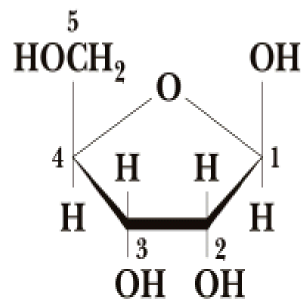
used in RNA used in DNA

The difference in str. of D-ribose & deoxyribose

- OH group on the 2-position of the sugar limits the range of possible secondary structures available to the RNA.
- Also makes it more susceptible to chemical and enzyme degradation

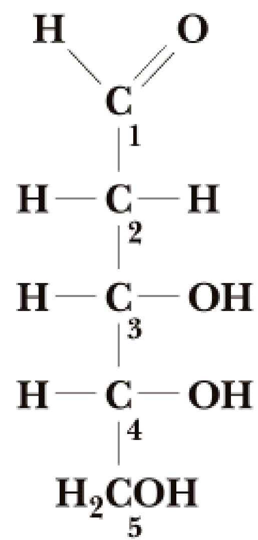


D-Ribose

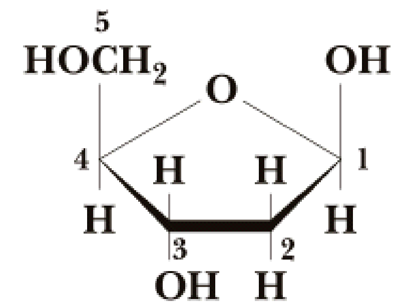


Furanose form of D-Ribose

β-D-Ribofuranose



D-2-Deoxyribose

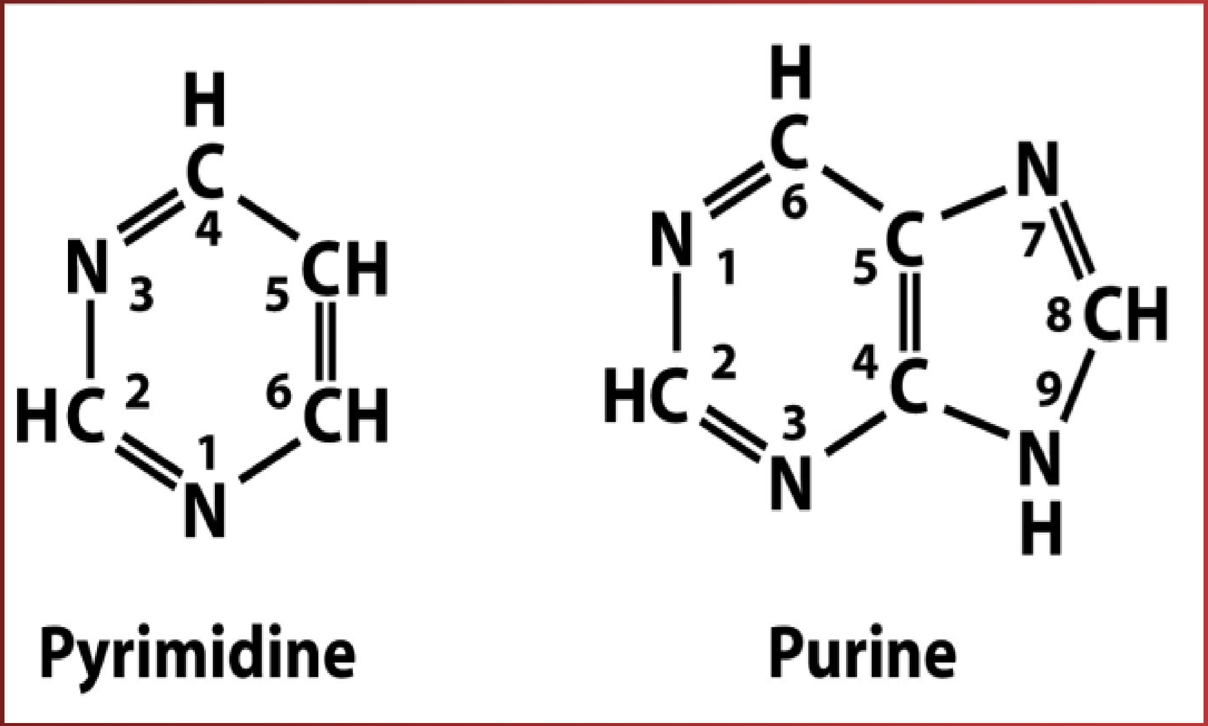


Furanose form of D-2-Deoxyribose

β-D-2-Deoxyribofuranose

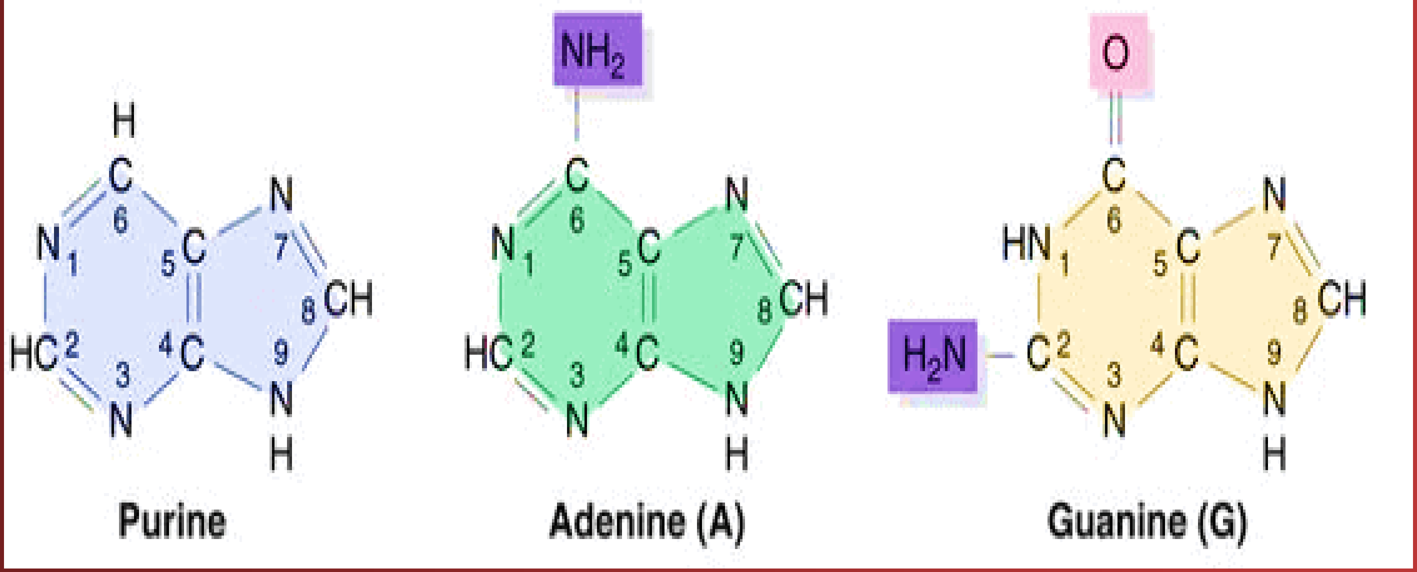
b- Nitrogenous bases

- Aromatic, heterocyclic molecules either Pyrimidine or a Purine derivative



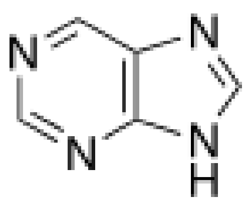
1- Purine bases

- Both types of nucleic acids contain the purine bases Adenine & Guanine, which contain two fused carbon-nitrogen rings.

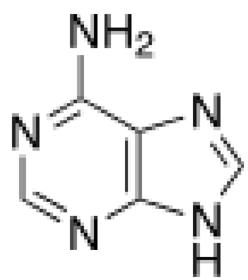


- There are other naturally occurring purine derivatives include Hypoxanthine, Xanthine, and Uric acid.

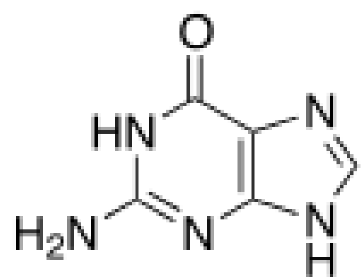
Naturally occurring purines



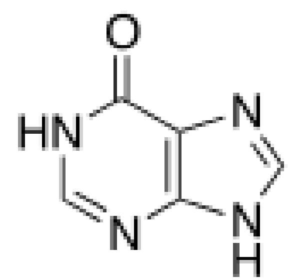
purine
1



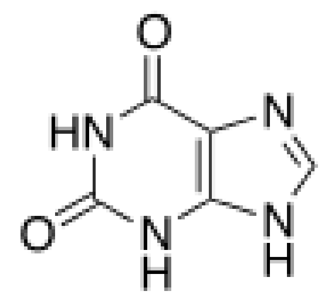
adenine
2



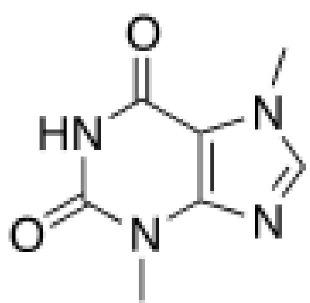
guanine
3



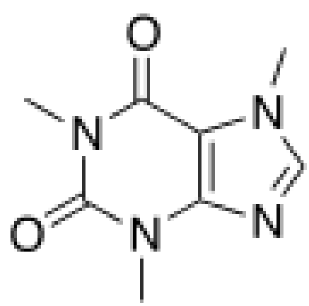
hypoxanthine
4



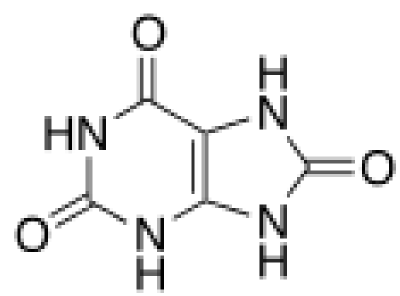
xanthine
5



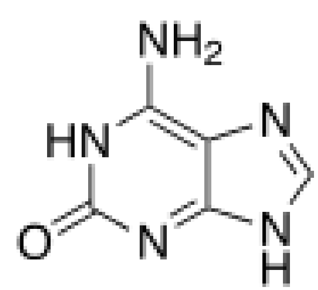
theobromine
6



caffeine
7



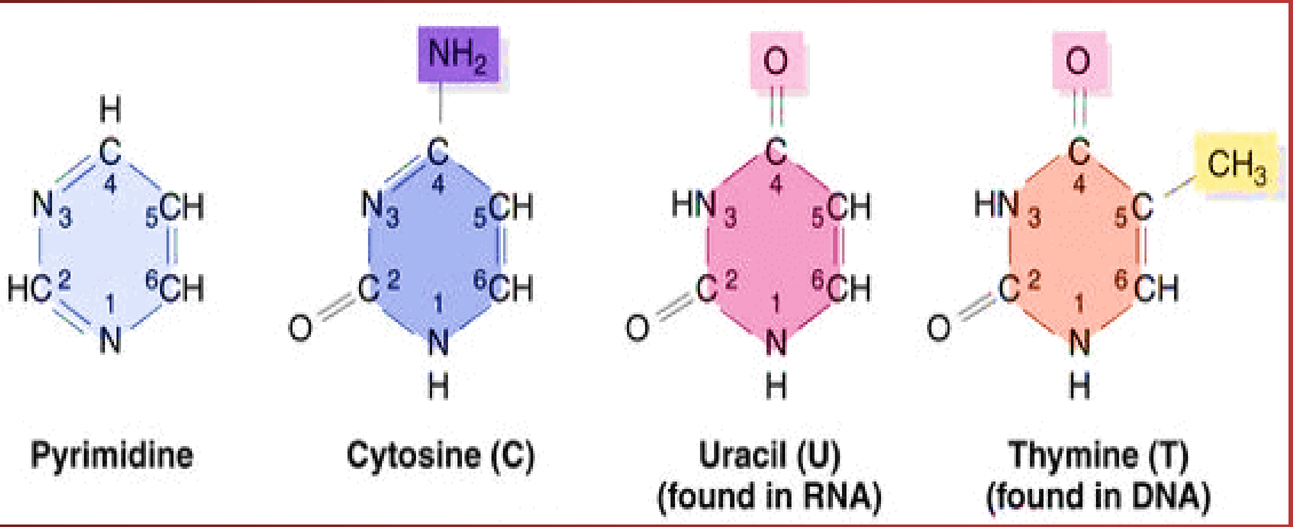
uric acid
8



isoguanine
9

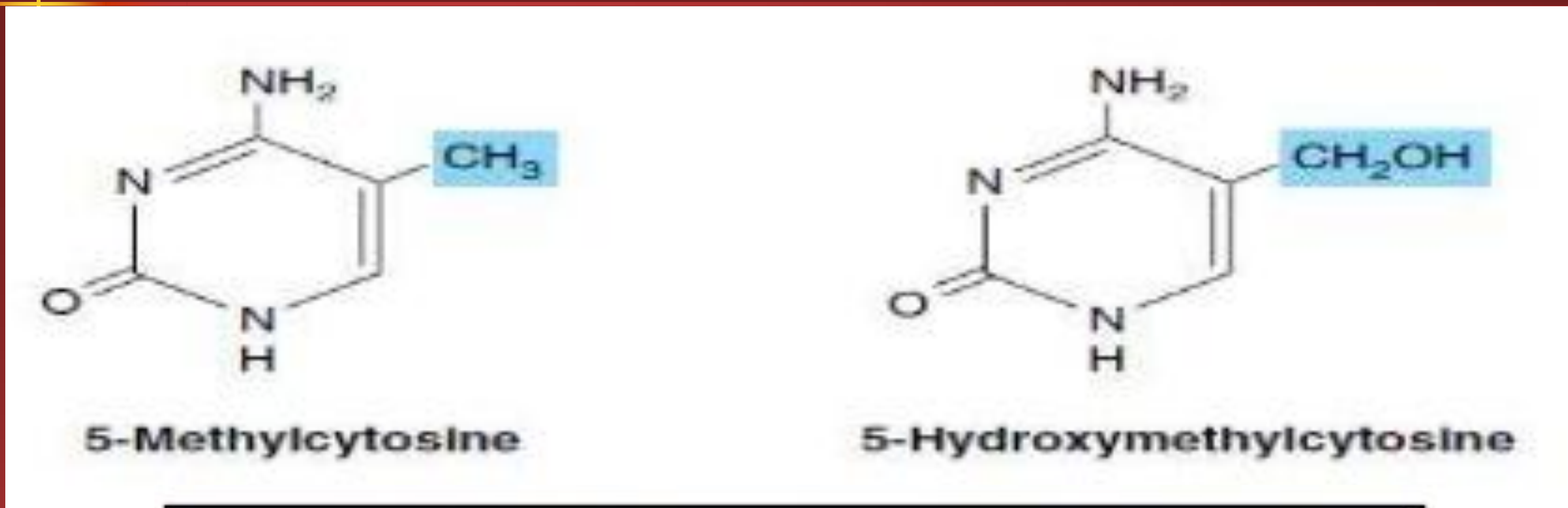
2- Pyrimidine bases

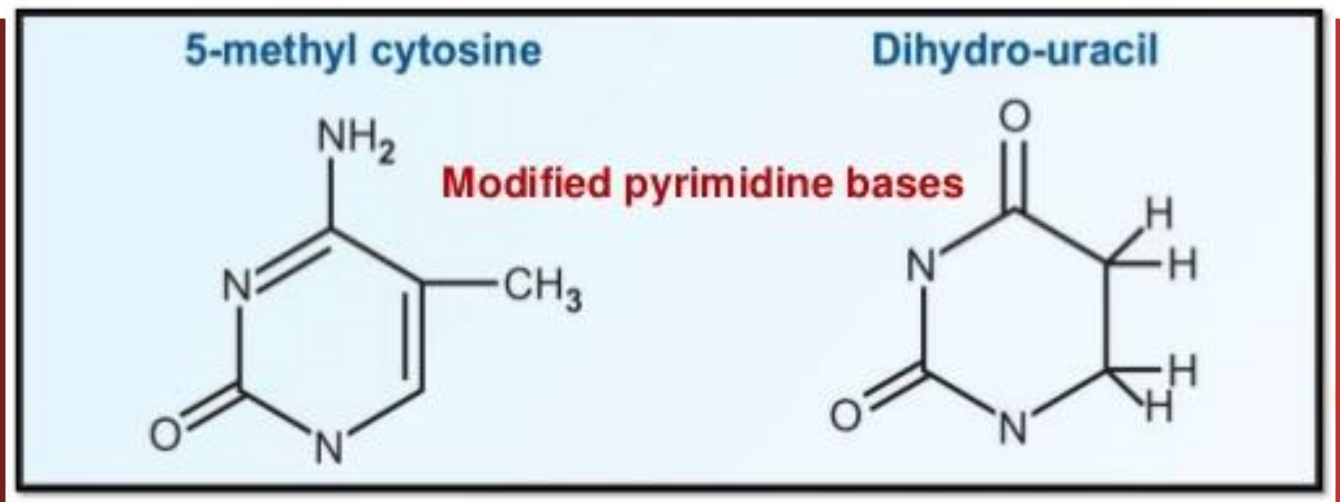
- Pyrimidine bases includes thymine, cytosine and uracil, all contain a single six carbon-nitrogen ring



- In certain viruses cytosine is replaced by 5-methylcytosine or 5-hydroxymethylcytosine

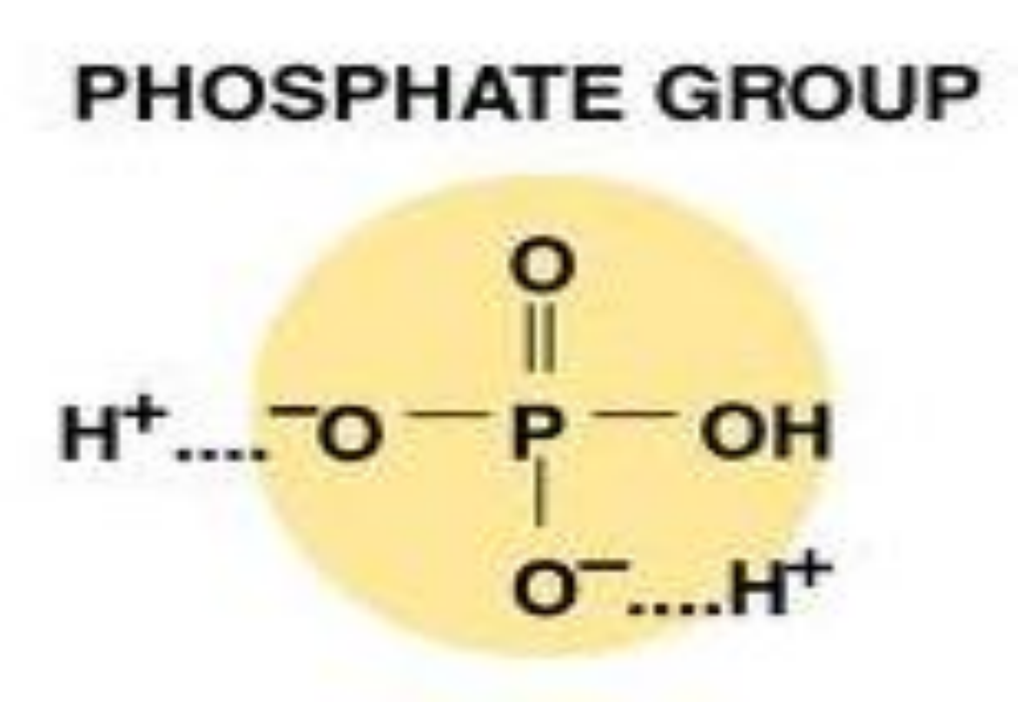
Modified Pyrimidine bases

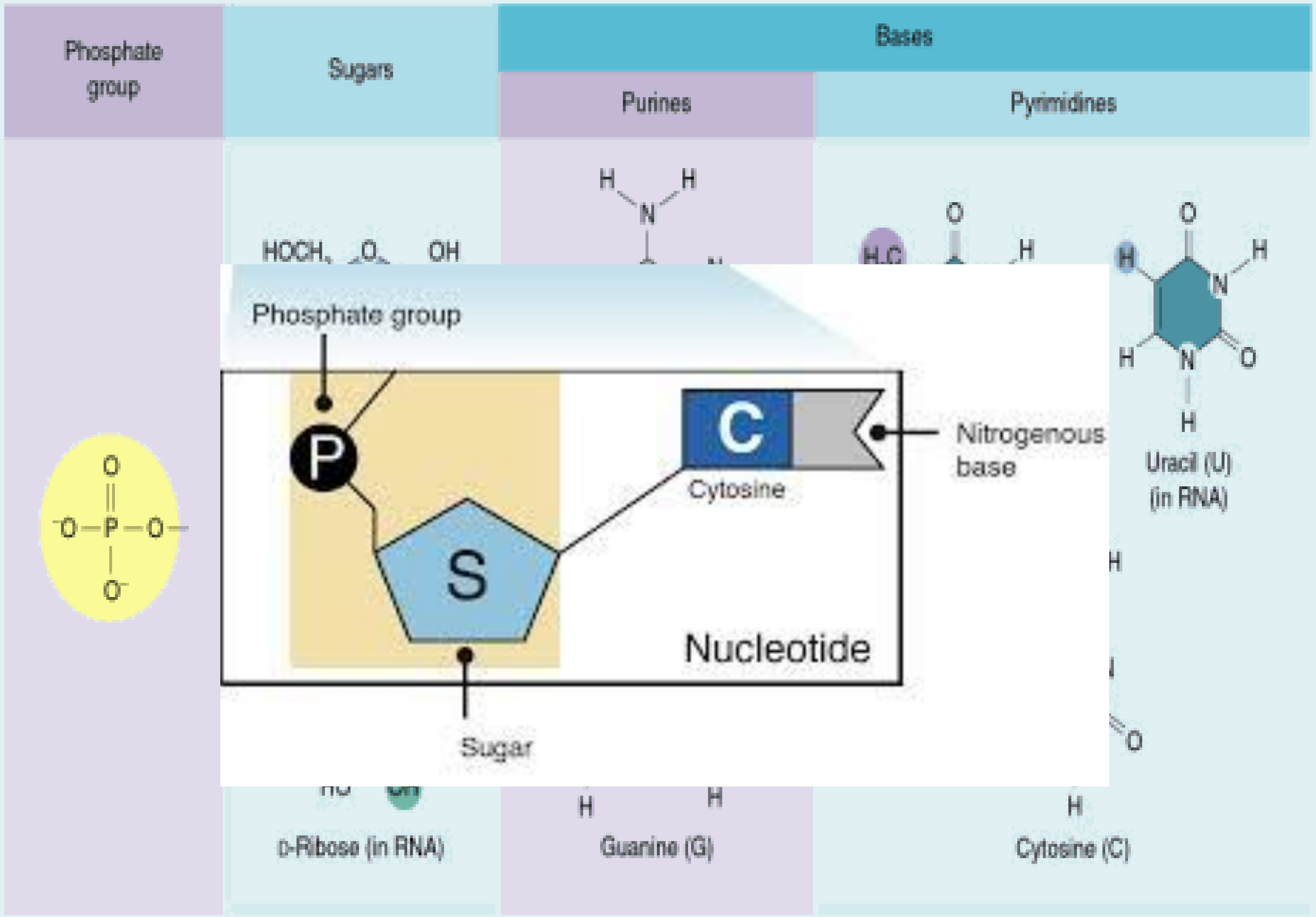




c- Phosphate group

- Molecule of Phosphoric acid





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